Name: _____

Class: _____

Date: _____ **Mr. Rawson • GCSE Physics**

Half-Term Reading – Changes of State and the Particle Model

Work through every reading in the tables; each blue title opens its page. Take notes on every reading in your notebook, in outline form, as shown on this page. Tick each specification statement when you can explain it without your notes.

Taking notes in outline form

How the outline works

- Give each reading its own heading: its title.
- **I, II, III** for each main idea on the page.
- **A, B, C** for the facts that explain that idea.
- **1, 2, 3** for examples and details.
- Write short phrases in your own words.
- Underline each key term the first time it appears.
- Finish each reading with a one-sentence summary.

Check: cover your notes, read only the Roman numerals, and explain each one out loud.

Two names, one idea. The Physics Classroom is an American site. Use the AQA words in your notes:

- $D = m/V \rightarrow \rho = m/V$
- g/mL or $g/cm^3 \rightarrow$ **kg/m^3** (1 g/cm^3 = 1000 kg/m^3)
- *vaporization* → **evaporating** or **boiling**
- *freezing*, *condensation*, *sublimation*: the same words

Lesson 1a: Solids, Liquids, and Gases The Physics Classroom

I. Three states of matter
   - A. told apart by what we observe and by how the particles behave
   - B. two more exist only in extreme conditions: plasma, Bose–Einstein condensate

II. Shape and volume
   - A. solid: fixed shape, fixed volume
     1. a pencil is the same in any room or container
   - B. liquid: shape of its container, fixed volume
     1. 250 mL of water is still 250 mL in a glass or a baking pan
   - C. gas: shape and volume of its container
     1. one breath fills the lungs (about 5 L), then a balloon (about 0.5 L)

III. The particle explanation
   - A. solid: strong attractions lock the particles in place
   - B. liquid: weaker attractions; particles flow past each other but stay together
   - C. gas: almost no attractions; particles spread out to fill the container

IV. Six changes of state
   - A. caused by heating, cooling or a change in pressure
   - B. melting, freezing, boiling (vaporising), condensing, subliming, deposition

Summary: solids, liquids and gases differ in shape and volume because the attractions between their particles are strong, weaker or almost absent.

When you finish: read only the box headings and the ticked statements, and explain each one out loud. Any statement you cannot tick yet, reread its BBC page.

Readings: The Physics Classroom (physicsclassroom.com); BBC Bitesize, GCSE Combined Science (AQA Trilogy). Specification statements: AQA GCSE Combined Science: Trilogy (8464), section 6.3.1. Links checked 5 October 2026. The worked outline and the particle diagrams are Mr. Rawson's.

P3.1 Changes of state and the particle model

AQA specification • P3.1 Changes of state and the particle model

P3.1.1 Density of materials

☐ **1** The density of a material is defined by the equation: density = $\frac{\text{mass}}{\text{volume}}$ $\rho = \frac{m}{V}$ density, ρ, in kilograms per metre cubed, kg/m^3; mass, m, in kilograms, kg; volume, V, in metres cubed, m^3. Students should be able to recall and apply this equation to changes where mass is conserved.

☐ **2** The particle model can be used to explain the different states of matter and differences in density.

☐ **3** Students should be able to recognise/draw simple diagrams to model the difference between solids, liquids and gases.

☐ **4** Students should be able to explain the differences in density between the different states of matter in terms of the arrangement of atoms or molecules.

P3.1.2 Changes of state

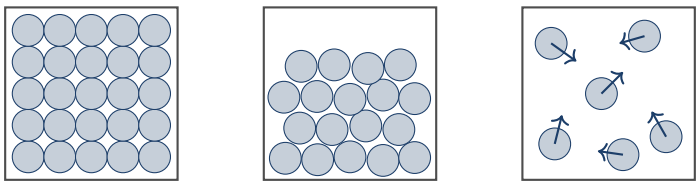
☐ **5** Students should be able to describe how, when substances change state (melt, freeze, boil, evaporate, condense or sublimate), mass is conserved.

☐ **6** Changes of state are physical changes which differ from chemical changes because the material recovers its original properties if the change is reversed.

Reading

Reading	**Covers** (statement numbers)
Matter 1a: Solids, Liquids, and Gases The Physics Classroom • *the example notes on page 1*	The particle arrangement in each state, and the changes between them, including sublimation (2, 3, 5)
Matter 2b: Chemical and Physical Changes The Physics Classroom	Why a change of state is a physical change (6)
Matter 4a: Concepts and Mathematics of Density The Physics Classroom	Density as mass per unit volume, and using $\rho = m/V$ (1)
Matter 4b: Measuring Density The Physics Classroom	Measuring the mass and volume of solids and liquids (1, Required Practical)
Density of materials: 1 Density BBC Bitesize (AQA)	The density equation in kg/m^3; why solids and liquids are denser than gases (1, 2, 4)
Density of materials: 2 Volume BBC Bitesize (AQA)	Volume of regular shapes, and of irregular ones by displacement (1)
Density of materials: 3 Investigating density experiment BBC Bitesize (AQA)	Required Practical 17: the method for solids and liquids
Temperature change and energy: 1 States of matter BBC Bitesize (AQA)	Particle diagrams; changes of state conserve mass and are reversible (3, 5, 6)

The three states, particle by particle (statements 2, 3 and 4)



solid **liquid** **gas**

	solid	**liquid**	**gas**
arrangement	regular pattern, touching	random, touching	random, far apart
movement	vibrate about fixed positions	move around each other	move quickly in all directions
density	high	high	low: the same mass takes up far more volume

Check: melt 1 kg of ice and you have 1 kg of water. The particles are rearranged, none are added or lost, so the mass is conserved and only the volume, and so the density, changes.